

R.U.T. LGT Toy-Derivation Falsifier Checklist

Reviewer-safe closure note for the Light-Geometry-Time toy expansion law

Core toy equation under review

The current LGT toy derivation should be judged against this minimal expansion law:

$$E_{\text{RUT}}^2(z) = E_{\Lambda\text{CDM}}^2(z) + \Omega_{\text{LGT},0} B(z) F_{\theta\omega}(z).$$

Minimal first-pass case:

$$E_{\text{RUT}}^2(z) = E_{\Lambda\text{CDM}}^2(z) + \Omega_{\text{LGT},0} B(z).$$

Scientific posture

This note does not present the LGT bridge as final proof. It defines the kill conditions that decide whether the toy bridge is doing real physical work or merely adding adjustable freedom.

Test / falsifier	Failure condition	Consequence
Shared-parameter transfer	No shared parameter region improves BAO while preserving growth/lensing behavior.	Reject or narrow the LGT bridge sector.
Physical amplitude	Required $\Omega_{\text{LGT},0}$ is unphysically large, unstable, or dominates the background where it should not.	Toy model fails as a healthy effective sector.
Universal bridge timing	$B(z)$ must turn on at unrelated epochs for different datasets.	Bridge is likely probe-specific fitting.
Mechanism dependency	Fit only works by moving r_d , H_0 , and Ω_m independently with no common LGT mechanism.	Derivation is not carrying explanatory weight.
Correspondence limit	$\Omega_{\text{LGT},0} \rightarrow 0$ does not recover $E_{\text{RUT}}^2(z) \rightarrow E_{\Lambda\text{CDM}}^2(z)$.	Model is structurally invalid.

Minimum next numerical test

Run a simple BAO toy scan using the minimal equation first. Only after that should the model add a nontrivial $F_{\theta\omega}(z)$. The first goal is not to win every dataset; it is to find whether one stable bridge region improves distances without breaking the standard limit.

Decision rule

Promising if the same LGT bridge parameters produce a stable improvement in BAO residuals, maintain a reasonable matter density, preserve the correspondence limit, and do not require growth/lensing to be sacrificed. Weak or failed if the bridge only works by retuning unrelated quantities separately.

Permanent reviewer-safe wording

The LGT toy derivation is a simulation-ready bridge scaffold, not a completed proof. Its value depends on whether the same light-flow/phase-order sector can transfer across BAO, growth, and lensing-sensitive observables without becoming probe-specific.